

EXTENDS AND IMPLEMENTS

1. C1 extends C2

Single Inheritance

Steps / Algorithm

1. Start the program.
2. Create a parent class C2.
3. Define a method in C2.
4. Create child class C1 using extends C2.
5. Define a method in C1.
6. Create an object of C1 in the main() method.
7. Call the methods of C2 and C1.
8. Stop the program.

Program

```
package extendsandimplements;
```

```
//Parent class
```

```
class Cat {
```

```
// Method of parent class
```

```
void show() {
```

```
    System.out.println("This is C2 class");
```

```
}
```

```
}
```

```
//Child class inherits C2
```

```
class Dog extends Cat {
```

```
// Method of child class
```

```
void display() {
```

```
    System.out.println("This is C1 class");
```

```
}  
}
```

//Main class

public class Extends {

// Main method

public static void main(String[] args) {

// Creating object of C1

Dog obj = new Dog();

// Calling parent class method

obj.show();

// Calling child class method

obj.display();

```
}
```

```
}
```

Output:

This is C2 class

This is C1 class

2. C1 extends C2, C3 ❌

Multiple Inheritance using Classes

Java does not support multiple inheritance using classes.

Steps

- 1. Start the program.**
- 2. Create two classes C2 and C3.**
- 3. Try to make C1 extend both classes.**
- 4. Java gives a compilation error.**
- 5. Therefore, C1 extends C2, C3 is not allowed.**

6. Stop the program.

Program

```
package extendsandimplements;
```

```
//First parent class
```

```
class C2 {
```

```
    void show() {
```

```
        System.out.println("C2 class");
```

```
    }
```

```
}
```

```
//Second parent class
```

```
class C3 {
```

```
    void display() {
```

```
        System.out.println("C3 class");
```

```
    }
```

```
}
```

```
//This is NOT allowed in Java
```

```
//class C1 extends C2, C3 {
```

```
//}
```

```
//Main class
```

```
public class ExtendsTwo {
```

```
    public static void main(String[] args) {
```

```
        // Display the reason
```

```
        System.out.println("A class cannot extend two classes in Java.");
```

```
}
```

```
}
```

Output

<terminated> Extends two [Java Application] C:\Users\GAYATHI

A class cannot extend two classes in Java.

3. C1 implements I1

Class Implementing an Interface

Steps / Algorithm

1. Start the program.
2. Create an interface I1.
3. Declare a method inside I1.
4. Create class C1.
5. Use implements I1.
6. Provide implementation for the interface method.
7. Create an object of C1.
8. Call the method.
9. Stop.

Program